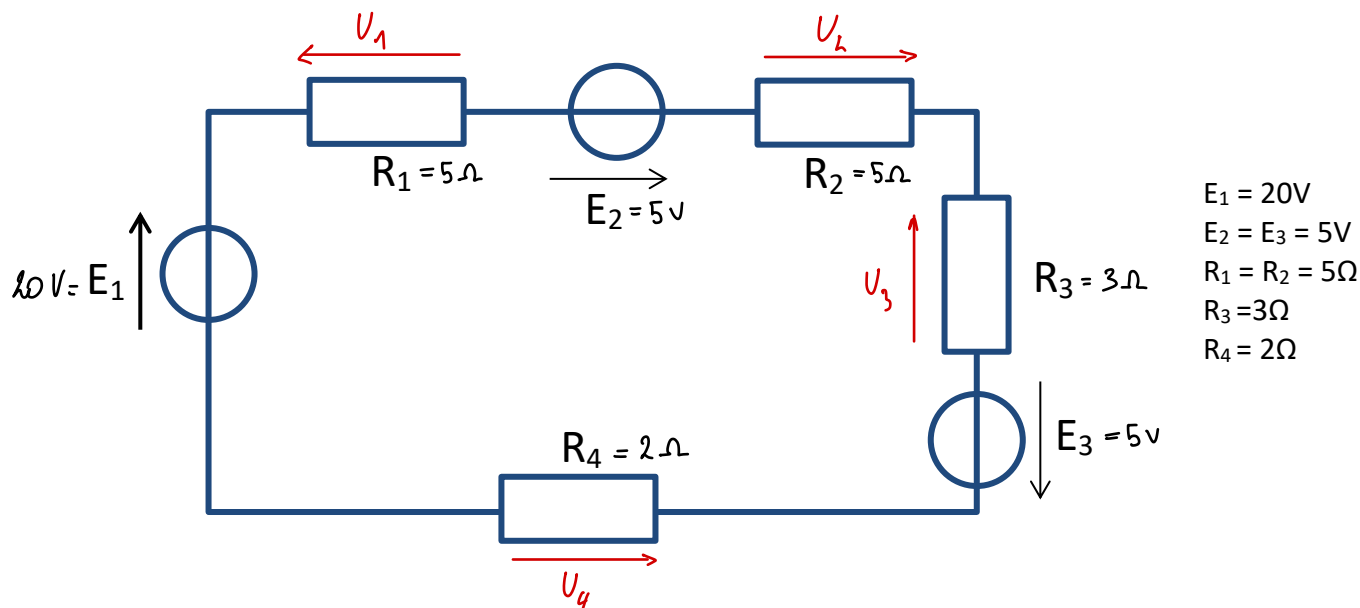


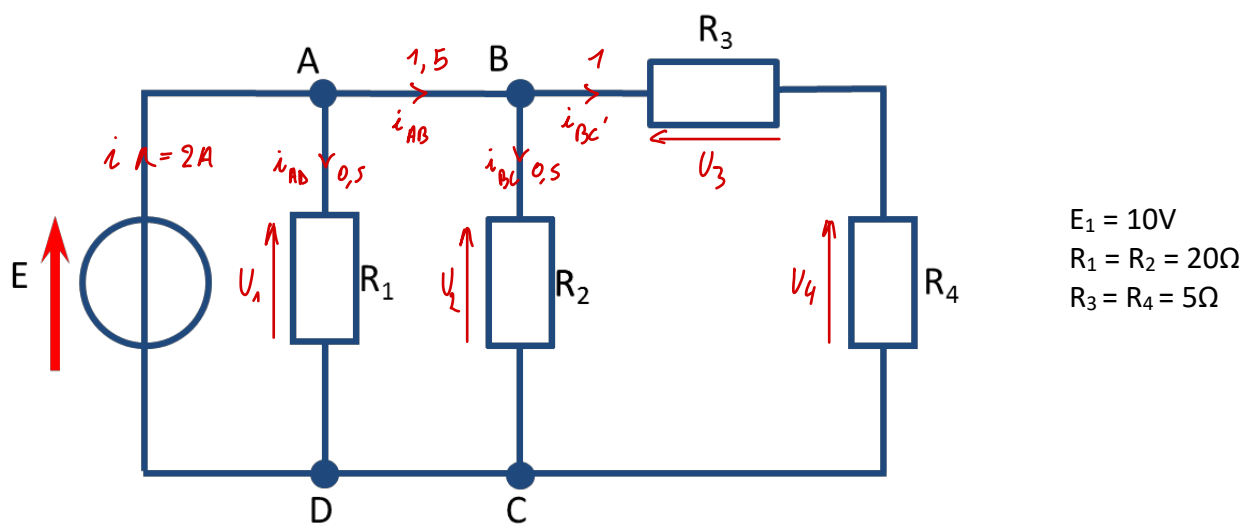
Compétences Visées :

- ☐ Proposer une démarche permettant de déterminer des grandeurs électriques.

Exercice 1 : Déterminer les intensités et tensions aux bornes de chaque composant du circuit ci-dessous :



Exercice 2 : Déterminer les grandeurs électriques (tension et intensité) dans chacune des branches des circuits ci-dessous :



Exercice 1

$$E_1 - U_1 + E_2 - U_2 - U_3 + E_3 - U_4 = 0$$

$$\Leftrightarrow E_1 + E_2 + E_3 - i(R_1 + R_2 + R_3 + R_4) = 0$$

$$\Leftrightarrow i = \frac{E_1 + E_2 + E_3}{R_1 + R_2 + R_3 + R_4} = \frac{20 + 5 + 5}{5 + 5 + 3 + 2} = 2A \quad \boxed{i = 2A}$$

$$\boxed{U_1 = R_1 \times i = 10V}$$

$$\boxed{U_2 = R_2 \times i = 10V}$$

$$\boxed{U_3 = R_3 \times i = 6V}$$

$$\boxed{U_4 = R_4 \times i = 4V}$$

Exercice 2:

Résistance équivalente: On a $R_1 // R_2 + R_3 + R_4$

$$\text{d'où } \frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{10} = \frac{1}{20} + \frac{1}{20} + \frac{1}{10} \quad R_{eq} = \frac{1}{\frac{1}{20} + \frac{1}{20} + \frac{1}{10}} = \frac{1}{0,05 + 0,05 + 0,1} = \frac{1}{0,2} = \frac{1}{\frac{1}{5}} = 5\Omega \quad \boxed{5\Omega}$$

$$U = Ri \Leftrightarrow E_1 = R_{eq} \times i \Leftrightarrow i = \frac{E_1}{R_{eq}} = \frac{10}{5} = 2A \quad \boxed{i = 2A} \quad \boxed{U_1 = U_2 = E_1 = 10V}$$

$$i_{AD} = \frac{U_1}{R_1} = \frac{10}{20} = 0,5A \quad \boxed{i_{AD} = 0,5A}$$

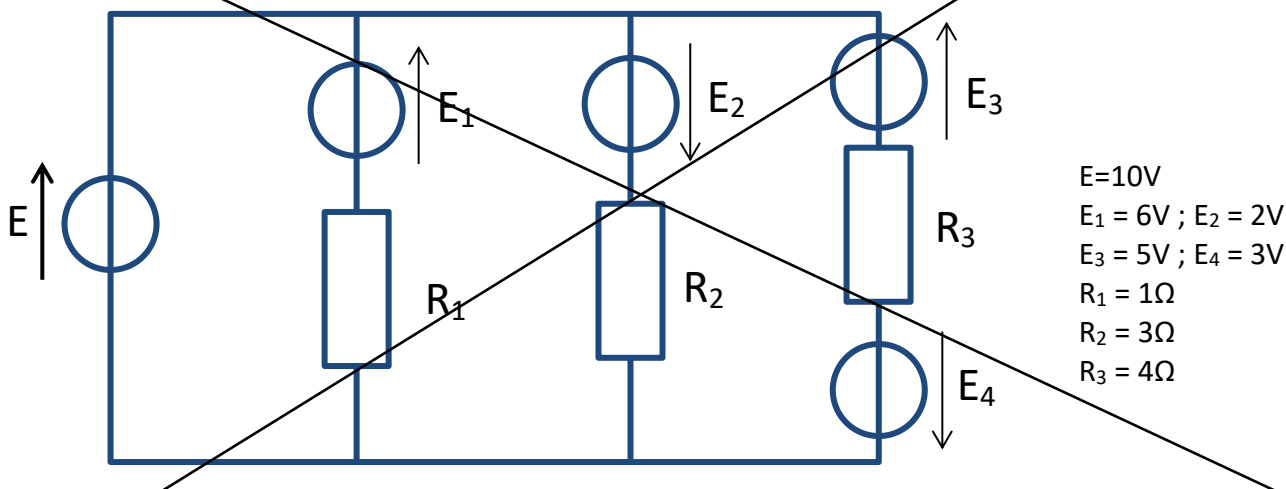
$$i_{BC} = \frac{U_2}{R_2} = \frac{10}{20} = 0,5A \quad \boxed{i_{BC} = 0,5A}$$

$$R_{eq1} = R_3 + R_4 = 10\Omega \quad i_{BC'} = \frac{U_{eq1}}{R_{eq1}} = \frac{10}{10} = 1A \quad \boxed{i_{BC'} = 1A}$$

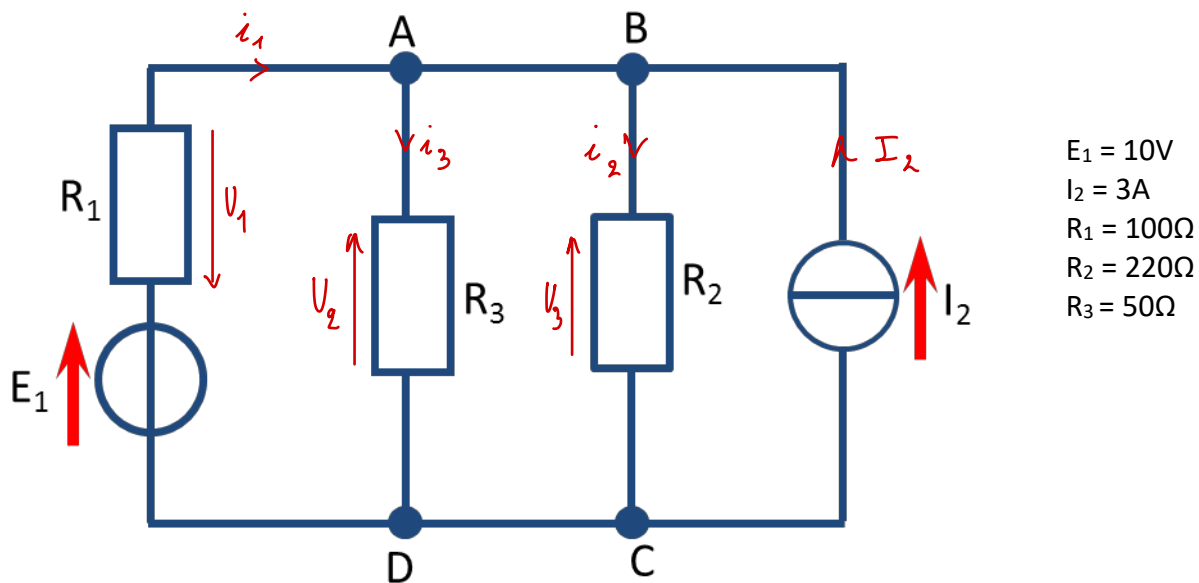
$$\text{loi des mailles } i = i_{AD} + i_{AB} \Leftrightarrow i_{AB} = 1,5A \quad \boxed{i_{AB} = 1,5A}$$

$$U_3 = U_4 = R_{3/4} \times i_{BC'} = 5 \times 1 = 5V \quad \boxed{U_3 = U_4 = 5V}$$

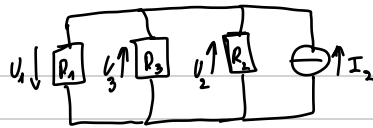
Exercice 3 : Déterminer l'intensité et le sens du courant I débité par la source E :



Exercice 4 : Déterminer les grandeurs électriques (tension et intensité) dans chacune des branches des circuits ci-dessous :



Exercice 3:



① On passive $E_1 \rightarrow \text{Fil}$

$$R_{eq} = \frac{R_1 R_2 R_3}{R_1 R_2 + R_1 R_3 + R_2 R_3} = \frac{100 \times 50 \times 220}{100 \times 220 + 100 \times 50 + 50 \times 220} = \frac{1100000}{38000} = \frac{1100}{38} \approx 29 \Omega$$

$$U_{I_2} = R_{eq} \times i_2 = 29 \times 3 = 87V$$

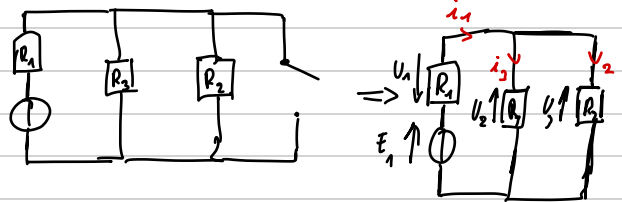
$$-U_1 = U_2 = U_3 = U_{I_2} = 87V$$

$$i_1 = \frac{U_1}{R_1} = \frac{-87}{100} = -0,87A$$

$$i_2 = \frac{U_2}{R_2} = \frac{87}{220} = 0,395A$$

$$i_3 = \frac{U_3}{R_3} = \frac{87}{50} = 1,74A$$

② On passive $I_2 \rightarrow \text{interrupteur ouvert}$



$$R_{eq} = R_1 + \frac{R_2 R_3}{R_2 + R_3} = 100 + \frac{220 \times 50}{220 + 50} = 100 + \frac{11000}{270} = 141 \Omega$$

$$U = R i \Leftrightarrow E_1 = R_{eq} \times i_1 \Leftrightarrow i_1 = \frac{E_1}{R_{eq}} = \frac{10}{141} = 0,07A \quad i_1 = 0,07A$$

$$E_1 = U_2 = U_3 = 10V$$

$$i_2 = \frac{U_2}{R_2} = \frac{10}{220} = 0,045A \quad i_2 = 0,045A$$

$$i_3 = \frac{U_3}{R_3} = \frac{10}{50} = 0,2A \quad i_3 = 0,2A$$

$$i_2 + i_3 \approx 0,245A$$

③ On somme :

$$i_1 = -0,87 + 0,07 = -0,80A$$

$$i_2 = 0,395 + 0,045 = 0,44A$$

$$i_3 = 1,74 + 0,2 = 1,94A$$

$$U_1 = 10 - 87 = -77V$$

$$U_2 = 10 + 87 = 97V$$

$$U_3 = 10 + 87 = 97V$$